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RESEARCH ARTICLE

Harvesting surface vegetation does not impede self-recovery of *Sphagnum* peatlands

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Ecosystem restoration frequently involves the reintroduction of plant material in the degraded ecosystem. When there are no plant nurseries or seeds available on the market, the plant material has to be harvested in the wild, in a "donor ecosystem." A comprehensive assessment of donor ecosystem recovery is lacking, especially for *Sphagnum*-dominated donor peatlands, where all top vegetation is harvested mechanically with different practices. We aimed to evaluate (1) the regeneration of vegetation, especially of *Sphagnum* mosses, to determine which harvesting practices are best to enhance recovery and (2) the influence of the site hydrological conditions and meteorological variables of the first complete growing season postharvesting on peat moss regeneration. Twenty-five donor sites covering a 17-year chronosequence (harvested 1–17 years ago) were inventoried along with 15 associated natural reference sites located in Quebec, New Brunswick, and Alberta, Canada. All donor sites aged 10 years or more were dominated by *Sphagnum* mosses, though plant composition varied between donor and their associated reference sites because of the wetter conditions at harvested donor sites. Harvesting practices strongly influenced donor site recovery, showing that the skills of the practitioner are an essential ingredient. Harvesting practices minimizing donor site disturbances are recommended, such as the choice of the adequate donor site (localization, hydrologic conditions, vegetation), the use of less disruptive methods, and harvesting when the soil is deeply frozen. This study demonstrated that harvesting surface plant material for peatland restoration is not detrimental towards the recovery of near-natural peatland ecosystems.

Key words: donor site, ecosystem restoration, harvesting practices, Moss Layer Transfer Technique, *Sphagnum*-dominated peatland

Implications for Practice

- Adequate harvesting practices and donor sites can speed up *Sphagnum* regeneration in the donor sites.
- Drier donor sites with a high cover of trees and ericaceous shrubs, forest mosses, and lichens should be avoided as a source of diaspores for peatland restoration. They provide low *Sphagnum* content and the recovery of these sites might be impeded by their drier nature.
- The disturbance of donor sites should be minimal. Best practices for harvesting would be avoiding very wet donor sites, and harvesting in the midsummer should be avoided when mosses are at their lowest regeneration potential and when the risk of machinery sinking is high.

Introduction

Ecosystem restoration often involves reintroduction of plants in order to reestablish ecosystem biodiversity and functions. Vascular plants can be reintroduced from seedlings cultivated in nurseries, from seeds or by planting resprouting plants (e.g. willows) harvested in the wild or from nurseries. However, for nonvascular plants, like bryophytes, moss nurseries are rare. Therefore, reintroduction material still mostly comes from natural ecosystems. The recovery of these "donor ecosystems" after vegetation harvesting for ecosystem restoration is raising concerns among the restoration ecologists and it was never assessed for peatland ecosystems.

In North America, the Moss Layer Transfer Technique (MLTT; Quinty & Rochefort 2003 and Graf et al. 2012 for a detailed description of the method) has been applied by the Canadian horticultural peat industry for more than 100 restoration projects for more than 1,100 ha across the country. This large-scale mechanized restoration technique is effective for reestablishing peatland vegetation (González & Rochefort 2014), hydrological attributes (Taylor et al. 2016), and restoring carbon sink function (Strack & Zuback 2013). The MLTT restoration approach includes the active reintroduction of typical peatland plant fragments (Graf et al. 2012). To do so, the top 10 cm of plant material (including mostly *Sphagnum* mosses, but also aerial and underground parts of vascular plants, seeds, and spores) is mechanically harvested in a nearby peatland.

Author contributions: MGN performed all the experimental preparation, fieldwork, analyzed the data, wrote, and edited the manuscript; SH, LR assisted in the development of the experimental design, helped with the experiment preparation, data analysis, and revised the manuscript.

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Peatlands used as donor sites are preferably sectors that are to be opened for peat extraction or machinery-accessible sectors of peatlands in the near-natural state and are reserved for this purpose in the peatland management plan of the peat industries.

Harvesting of plant fragments and other diaspores ideally occurs when the conditions are not too wet. Harvesting methods generally consist of shredding the surface vegetation (more commonly with a rototiller, but a rotary harrow or a forest mulcher, have been used) and collecting the fragments with an excavator or bulldozer. Harvesting plant material for restoration is done at a ratio of 1:10, meaning that the donor site is 10 times smaller than the restored site (Quinty & Rochefort 2003; Graf et al. 2012).

Donor site regeneration raises concerns among environmental communities, governments, and certification agencies. There is a need to evaluate the impact of restoration; in other words, is it worth it to restore a peatland if the donor site (a natural peatland ecosystem) is destroyed? This is especially important in areas where pristine peatlands are scarce and often legally preserved, and where finding peatland plant material for reintroduction becomes a challenge. If donor site regeneration is not assured, should we find other sources of plant material (culture in greenhouses or in outdoor farming settings) and forget the approach of harvesting in natural peatlands? Are the conservation of natural peatlands and peatland restoration activities complementary or opposed?

Worldwide, an increasing number of policies from environmental and climate agencies are requiring more responsible and sustainable management practices for peatlands affected by human activities (e.g. United Nations Framework Convention on Climate Change, Reducing Emissions from Deforestation and Forest Degradation, EU Habitats Directive; Stoneman et al. 2016). Another example, the Veriflora certification, prevalent in the North American horticultural and floral industries, encourages that all peat fields where peat extraction activities recently ceased should be restored with the MLTT and that donor sites should be set aside during the peatland development plan (SCS Global Services 2012). Outside North America, there is also an increasing demand for *Sphagnum* propagules as plant reintroduction material within the MLTT (e.g. in Estonia—Karofeld et al. 2016; Lithuania—Sendzikaite et al. 2017; United Kingdom—Peacock et al. 2013; Australia—Whinam et al. 2010; Chile—Domínguez 2014) or for the implantation of *Sphagnum* farms (in Canada—Pouliot et al. 2015; Germany—Gaudig et al. 2017; Japan—Hoshi 2017). The demand for little-decomposed *Sphagnum* fibers for orchid propagation or plant packaging market is also increasing (Zegers et al. 2006) for which top surface *Sphagnum* fibers are harvested in natural peatlands (New Zealand—Whinam & Buxton 1996; Finland—Silvan et al. 2017; Chile—Díaz & Silva 2012). Demand for little-decomposed *Sphagnum* fibers translates into pressure and environmental impacts on natural peatlands and their recovery after harvesting needs to be investigated as well as the influence of harvesting practices and environmental factors.

Hydrology was pointed out as the main environmental factor controlling *Sphagnum* regeneration in natural hand-harvested peatlands, for example, in Chile (Díaz & Silva 2012) and New Zealand (Whinam & Buxton 1996). According to Whinam and Buxton (1996), selected peatlands for harvesting should have a stable water table level, not too high nor too low, but they do not refer to a specific level. The same authors recommended that the microtopography at the surface of the donor site after plant harvesting should ideally be even to optimize water distribution and drainage. Also, according to them, the use of machinery should be minimized or avoided because it causes ruts, irregular water distribution, and exposes more humified peat resulting in slowed or absent *Sphagnum* regeneration. Lastly, the presence of leftover *Sphagnum* fragments (approximately 30% of cover) fostered recovery in natural hand-harvested peatlands (Whinam & Buxton 1996). The landscape context of machinery-harvested plant material for reintroduction in North America provides postharvested peatland conditions that differ from hand-harvested peatlands and requires a comprehensive evaluation of their recovery.

This study is the first to evaluate regeneration of donor sites at a broader geographical scale including 25 donor sites across Canada. We aimed to (1) evaluate the regeneration of vegetation (*Sphagnum* mosses, other cryptogams [mosses and lichens] and vascular plants) in terms of abundance and composition, (2) determine the impact of harvesting practices on the regeneration of vegetation, and (3) evaluate the influence of the site hydrology and meteorology of the first-year post plant material harvesting on the regeneration of *Sphagnum* mosses.

Methods

Study Area

This study was conducted on 25 peatlands that were used as donor sites for *Sphagnum*-dominated peatland restoration with the Moss Layer Transfer Technique. The donor sites were located in the province of Quebec, New Brunswick (eastern Canada), and Alberta (western Canada) and were monitored for 1–17 years postharvesting (Table S1, Supporting Information; Fig. 1). Sites are located in four major Canadian climatic regions: the Northeastern Forest Region, the Great Lakes and St. Lawrence Lowlands Region, the Atlantic Region, and the Northwestern Forest Region (Table S1) (Environment and Climate Change Canada 2016). These peatlands are mainly rain-fed open bogs (National Wetlands Working Group 1997) dominated by *Sphagnum* mosses and ericaceous shrubs.

Donor sites were selected by peatland managers and supposed to contain at least 50% ground cover of *Sphagnum* mosses from the *Acutifolia* subgenus (mainly *Sphagnum fuscum*, *Sphagnum rubellum*) and *Polytrichum strictum* (Quinty & Rochefort 2003; Graf et al. 2012). Donor sites are usually chosen among suitable ones to minimize the distance to the restoration sites to reduce transportation costs. Therefore, donor sites are often surrounded by or adjacent to drained peat fields. Because different sectors of an extracted peatland can be restored in different years, there were 25 surveyed donor sites located in 15 peatland sites/industrial entities (Table S1).

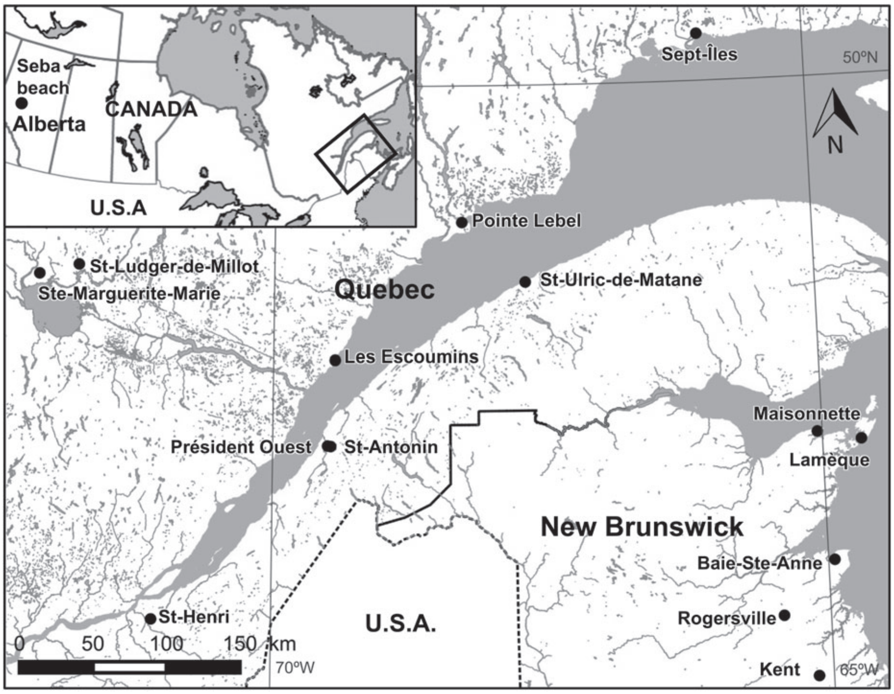


Figure 1. Location of the 15 peatlands with donor sites in the provinces of Quebec, New Brunswick, and Alberta (Canada).

Site Description

A major constraint to donor site selection for this ecological assessment was that some sites no longer existed at the time of this study because they have been opened for peat extraction. Therefore, the assessment does not represent an age continuum and is not replicated. To avoid misinterpretation based on low effective sample size per age, the donor sites were classified into three categories: 5 years or less ($n = 12$), between 5 and 10 years ($n = 6$), and 10 years or more since harvesting ($n = 7$).

Natural reference sites represent the vegetation composition of the donor sites before collection of the diaspores and act as a standard comparison for donor site recovery. One reference site per peatland site was selected, so a given reference site could be the reference for several donor sites. Sometimes, when all of the open parts of the *Sphagnum* peatland were harvested, there was no adjacent natural *Sphagnum* peatland to serve as a reference. In these few cases ($n = 5$), the reference site was still just aside the donor site but corresponded more to a treed *Sphagnum* peatland (Fig. S1).

Assessment Approach

Vegetation Inventories. The postharvesting surveys were performed in 40 sites (25 donor sites and 15 reference sites) and documented the plant communities in 141 plots (5×5 m) (Table S1). Two to four plots were surveyed per donor site or reference site depending on the size of the site and heterogeneity of regeneration. Plots were located randomly and distributed within each site to represent the site heterogeneity. In each donor site or reference site, vegetation was surveyed either during summer 2015 or 2016. Donor site location was sometimes difficult to determine, especially for older sites. A set of GPS points provided by the peatland managers was used to find the general location of the site and the specific perimeter of the harvest was determined using physical parameters of the vegetation (such as a uniform lower height of the ericaceous shrubs stratum).

The total cover for all the vegetation, peat and litter, and disturbed substrate were estimated by their vertical projections on the ground in the 5×5–m plots. Total vascular plant cover and cover by species were evaluated within four quadrats (1×1 m), while total bryophytes cover, total *Sphagnum* cover, and cover by species were recorded in 12 quadrats (25×25 cm) that were systematically distributed within the plot. In addition, the total cover of trees and shrubs, ericaceous shrubs, graminoid plants, forbs, other mosses than *Sphagnum*, lichens, and liverworts were calculated by adding the mean cover of each species corresponding to these strata, and therefore included species superposition.

Nomenclature follows VASCAN, the Database of Vascular Plants of Canada (Brouillet et al. 2010), Flora of North America Editorial Committee (1993+) for bryophytes and Lichens of North America (Brodo et al. 2001) for lichens.

Harvesting and Environmental Conditions. Explanatory variables that could influence donor site regeneration were classified into three groups, harvesting, hydrological proxies, and selected meteorological variables or indices known to be particularly meaningful to the regeneration of mosses (González & Rochefort 2014; Table 1). Some explanatory variables were measured at the plot level, others at the donor site level.

Table 1. Description and units of the explanatory variables measured (harvesting, hydrology, meteorology).

Group	*Name*	*Description*	*Notes*
Harvesting	Time	Time since harvesting (year)	The number of complete growing seasons since the harvesting was provided by the peatland managers.
	Shape	The shape of the area harvested (two classes: rectangle or square)	Long and narrow (rectangle) donor sites are supposed to facilitate the work with the machinery and minimize sinking (Quinty & Rochefort 2003).
	Method	Harvesting method (four classes: rotary harrow, mulcher, rototiller, excavator with a large bucket with comb teeth)	The rotary harrow, mulcher, and rototiller were pulled by a tractor within the donor site, while the excavator with a large bucket with comb teeth operated from outside the donor site.
	Season	Harvesting season (four classes: spring, summer, fall, winter)	The season was not used to represent a phenological variable but rather a reflection of the different levels of site disturbance caused by the machinery during harvesting. For example, there is a greater likelihood for the machinery to cause ruts in the summer because there is no ice underneath the peat to support the machinery.
	Natural, extracted restored, unrestored, anthropic	Relative use of the 500-m buffer zone around the donor site (%)	Using satellite images from Google Earth, a buffer zone of 500 m around the donor sites was determined in which the relative areas (%) of the following uses was calculated: natural (mainly forest, water bodies, and peatlands), current peat extraction, previously extracted and restored peatland, previously extracted and not restored peatland, and anthropic land uses other than peat extraction (mainly agriculture). Satellite images from the first or second year following the harvesting were used.
Hydrological	Dist ditch	Distance to the closest drainage ditch (two classes: <15 m, >15 m)	Hydrology at each plot was evaluated using two hydrological proxies. Proxies were used because donor sites were not monitored at the same period of the summer nor at the same year. Implementing a hydrology monitoring program with water table depth measures during the growing season was too expensive and intensive to put in place, due to site remoteness and the number of sites. By photointerpretation of satellite images (Google Earth) from the first or second year following the harvesting and validation on the field, the approximate distance to the closest drainage ditch and its status was determined for each plot.
	Ditch type	Type of the closest drainage ditch (two classes: open active, blocked)	
Meteorological	Pspr, Psum, Pfall	Cumulative precipitations (mm)	Weather data for each site was retrieved from the closest meteorological station (Environment Canada). Only data from the first complete growing season was used because weather during subsequent years had a minor effect on *Sphagnum* establishment (Chirino et al. 2006).
	Tspr, Tsum, Tfall	Mean temperature (°C)	Cumulative precipitations and mean temperature for spring (May, June), summer (July, August), and fall (September, October) were calculated.
	Per rain	Percentage of rainy days (%)	Total percentage of rainy days and the maximum number of consecutive days without precipitations (0 mm) during the summer were calculated.
	Max noP	Maximum number of days without rain	
	AI	Aridity index	Aridity index equals the annual precipitation falling as rain divided by the annual mean temperature plus 10 (Gignac & Vitt 1990).

Statistical Analyses

Regeneration of the Vegetation. First, a linear regression was run to evaluate the effect of the number of years postharvesting on the total cover of *Sphagnum* mosses in the donor sites. Means of total *Sphagnum* cover per plot were used (donor sites: $n = 90$, natural reference: $n = 51$).

Second, absolute cover values for each vegetation layer (vascular plants, trees and shrubs, ericaceous shrubs, graminoid plants, forbs, bryophytes, *Sphagnum* mosses, other mosses than *Sphagnum*, lichens, liverworts, peat and litter, disturbed substrate) were averaged per site to generate a vegetation stratum matrix with one row per site (donor and reference) and one column per stratum (dimensions: 40×12). Stratum cover data were transformed using the Hellinger transformation method prior to analysis to avoid the double-zero problem (Legendre & Gallagher 2001). The Hellinger-transformed vegetation stratum matrix was subjected to a *k*-means partitioning. This analysis allowed us to group the observations into *k* groups that could be interpreted in terms of vegetation regeneration. Simple structure index (SSI) criterion was used to determine the appropriate number of clusters. This criterion was used instead of the Calinski–Harabasz criterion (recommended by Milligan (1996)) because it suggested a more interpretable number of clusters. Then, the distribution of each type of site (<5 years, 5–10 years, >10 years, and reference) was calculated in each *k* group. The *k*-means partitioning results were visualized using a principal component analysis (PCA) biplot using the same vegetation matrix where a circle of equilibrium contribution was drawn to determine which strata contributes significantly to the PCA.

Third, cover values for each species (vascular plants and cryptogams) were averaged per plot to generate a species matrix with one row per plot (donor and reference) and one column per species (dimensions: 141×52). This species matrix was also transformed using the Hellinger transformation. Only species of a frequency and/or a mean abundance of more than 5% were included to minimize the bias induced by rare species. An indicator species analysis (IndVal; Dufrêne & Legendre 1997) was run on the Hellinger transformed species matrix to identify indicator species in each previously determined *k*-means group. This analysis calculates measures of specificity and fidelity for each species to each group. Afterward, it calculates an IndVal value for each species that ranges from 0 to 1; when a value is close to 1, this species is considered to be an indicator and specific to a certain group. The signification of the indicator value of each species was evaluated with a randomization procedure with 9,999 permutations (Legendre & Legendre 2012).

Finally, to give a general portrait, descriptive analyses of the vegetation data of the previously determined *k*-means groups were done.

Influence of Harvesting and Environmental Conditions on Vegetation. Using only donor sites, cover values for previously selected vegetation layer by the circle of equilibrium contribution per plot were used to generate a vegetation stratum matrix with one row per plot and one column per stratum (dimensions: 90×5). Stratum cover data were transformed using Hellinger transformation method prior to analysis to avoid the double-zero problem (Legendre & Gallagher 2001).

To evaluate the role of each explanatory group (harvesting, hydrology, and meteorology) on vegetation regeneration, a redundancy analysis (RDA) was carried out on the vegetation stratum matrix per plot. Because collinearity between explanatory variables was expected, a forward selection was implemented before running the RDA. Explanatory variables that were highly correlated (Spearman coefficient > 0.7) were removed. The RDA was run with the selected significant explanatory variables. The significance of the RDA was tested with a permutation test using 9,999 randomized runs (Legendre & Legendre 2012). In this analysis, the experimental unit considered is the plot, therefore this might be considered as pseudoreplication (repeated measurements within a donor site). A partial RDA approach using site as a random factor was not used, because that way the influence of explanatory variables on the stratum matrix will be underestimated. Indeed, many explanatory variables (from harvesting and meteorology) were measured at the donor site level, not at the plot level. By including a random site effect, variance linked to these explanatory variables would have been lost. Finally, to distinguish the effect of time from the other harvesting variables and the relative contribution of all previously selected explanatory variables, a variation partitioning analysis was carried out on the vegetation stratum matrix.

All analyses were conducted with R (R Core Team 2016, version 3.3.1) software, more precisely decostand, cascadeKM, ordistep, rda, RsquareAdj, and varpart from vegan package (Oksanen et al. 2016), ordiequilibriumcircle from BiodiversityR (Kindt & Coe 2005), and indval from labdsv (Roberts 2016).

Results

Regeneration of Vegetation

Sphagnum cover in donor sites linearly increased with the number of years since harvesting (Fig. 2; adjusted $r^2 = 0.66$, $F_{[1,13]} = 15.33$, $p < 0.001$). According to the regression equation (Fig. 2), a *Sphagnum* cover of 70%, which is the mean value in reference sites, could be obtained in approximately 11 years.

Donor sites and reference sites were grouped in the ordination diagram (Fig. 3) into four groups that were defined in terms of vegetation regeneration: bare and disturbed, in regeneration, wet *Sphagnum*, and dry *Sphagnum* (Figs. 3 & S2). Only the cover of peat and litter, disturbed substrate, graminoid plants, *Sphagnum* mosses, and ericaceous shrubs contributed significantly to the ordination.

The bare and disturbed group ($n = 4$) was characterized with a low cover of living vegetation (approximately 15%), a high cover of bare peat and litter (approximately 85%) with still visible disturbed substrate (approximately 75%) (Table 2), and comprised exclusively sites of less than 5 years old (Fig. 3B). *Eriophorum vaginatum* and *Betula* sp. were indicators for this group (Table 3; IndVal = 0.54, $p = 0.02$ and IndVal = 0.27, $p = 0.03$, respectively). The in regeneration group ($n = 12$)

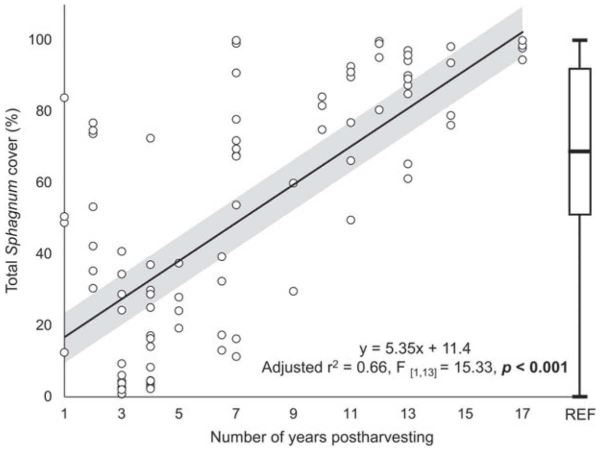
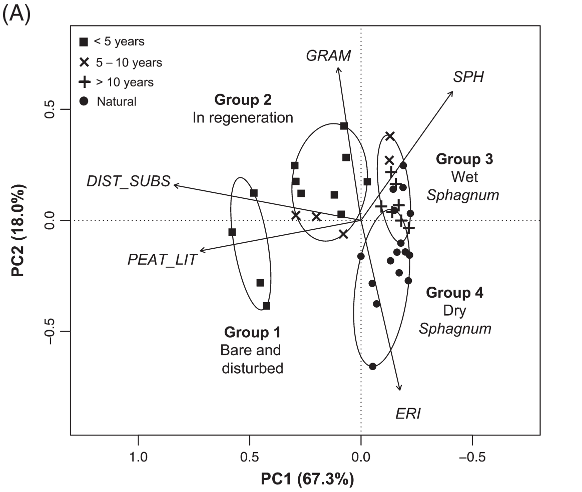


Figure 2. Relation between the total *Sphagnum* cover (%) (mean ± $CI_{95\%}$ [gray area]) per plot ($n = 90$) according to the number of years postharvesting in donor sites. Total *Sphagnum* cover values found in natural reference sites (REF) are represented in a boxplot ($n = 51$).

(A)



(B)

Group	< 5 years	5 – 10 years	> 10 years	Reference
1 – Bare and disturbed	33	0	0	0
2 – In regeneration	67	60	0	0
3 – Wet *Sphagnum*	0	40	100	25
4 – Dry *Sphagnum*	0	0	0	75

Figure 3. (A) Principal component analysis (PCA) biplot (scaling 1) of the total cover of the following strata: PEAT_LIT = peat and litter, DIST_SUBS = disturbed substrate, GRAM = graminoid plants, SPH = *Sphagnum* mosses, and ERI = ericaceous species. The significant strata were selected with the equilibrium circle procedure (not shown). The groups obtained by *k*-means partitioning are represented by ellipses (bare and disturbed, in regeneration, wet *Sphagnum*, and dry *Sphagnum*). (B) Distribution (%) of each type of sites (<5 years, 5–10 years, >10 years, and natural reference) in each of the groups of the *k*-means partitioning done on the vegetation stratum matrix (bare and disturbed, in regeneration, wet *Sphagnum*, and dry *Sphagnum*).

showed an intermediate cover of *Sphagnum* mosses (approximately 30%) and the highest cover of graminoid plants of all four groups (approximately 15%), of which *E. vaginatum* was the dominant species (Table 2). Most of the sites aged of 5 years and less and sites between 5 and 10 years old were in this group (Fig. 3B). Unidentified *Sphagnum* mosses in regeneration were strong indicators of the group (Table 3; IndVal = 0.70, $p = 0.01$). The wet *Sphagnum* group ($n = 12$) was described by the highest cover of *Sphagnum* mosses (approximately 85%) characterized with a high proportion of

Table 2. Group names and percentage of cover (%) of the most abundant strata and species (mean ± standard error [SE]) in each of the groups of the *k*-means partitioning (bare and disturbed, in regeneration, wet *Sphagnum*, and dry *Sphagnum*). An asterisk indicates a cover lower than 1%.

	1-Bare and disturbed	*2-In regeneration*	*3-Wet* Sphagnum	*4-Dry* Sphagnum
Total vegetation	17 ± 2	56 ± 3	94 ± 2	89 ± 2
Peat and litter	85 ± 4	49 ± 4	9 ± 1	13 ± 3
Disturbed substrate	76 ± 8	31 ± 5	2 ± 0.5	2 ± 2
Trees and shrubs	*	*	2 ± 0.4	5 ± 1
Betula papyrifera	*	*	*	*
Larix laricina	*	*	1 ± 0.4	*
Picea mariana	*	*	*	3 ± 0.9
Ilex mucronatus	*	0	0	*
Ericaceous shrubs	12 ± 3	16 ± 2	25 ± 2	57 ± 3
Chamaedaphne calyculata	4 ± 0.6	7 ± 0.9	9 ± 0.9	10 ± 2
Kalmia angustifolia	5 ± 2	4 ± 1	6 ± 1	25 ± 4
Rhododendron groenlandicum	1 ± 0.3	2 ± 0.4	5 ± 1	11 ± 2
Herbaceous species	3 ± 0.9	16 ± 2	7 ± 1	4 ± 0.7
Eriophorum vaginatum	1 ± 0.6	11 ± 2	4 ± 1	*
Sphagnum mosses	6 ± 2	33 ±4	85 ± 3	60 ± 5
Sphagnum angustifolium	*	3 ± 1	5 ± 2	4 ± 2
Sphagnum fuscum	2 ± 0.5	3 ± 1	10 ± 4	23 ± 5
Sphagnum magellanicum	*	6 ± 2	10 ± 2	3 ± 1
Sphagnum rubellum	5 ± 1	15 ± 3	51 ± 5	21 ± 4
Other cryptogams	4 ± 0.6	12 ± 2	8 ± 1	19 ± 3
Polytrichum strictum	3 ± 0.5	6 ± 1	5 ± 1	4 ± 1
Cladonia sp.	*	1 ± 0.4	*	10 ± 3

Table 3. Indicator species, IndVal, *p*-value, and frequency (number of occurrences) in each of the groups of the *k*-means partitioning done on the vegetation stratum matrix (bare and disturbed, in regeneration, wet *Sphagnum*, and dry *Sphagnum*).

Group	*Species*	*IndVal*	p	*Frequency*
1—Bare and disturbed	*Eriophorum vaginatum*	0.54	0.02	8
	Betula sp.	0.27	0.03	6
2—In regeneration	Unidentified *Sphagnum*	0.70	0.01	3
4—Dry *Sphagnum*	*Picea mariana*	0.63	0.007	13
	Rhododendron groenlandicum	0.60	0.006	14
	Kalmia angustifolia	0.58	0.001	31

wetter microhabitat species (*S. angustifolium, S. rubellum, S. magellanicum*) compared to the next group that had a higher proportion of hummock species (Table 2). *Chamaedaphne calyculata* was the only ericaceous species reaching similar cover than in the next group (Table 2). All donor sites aged more than 10 years old were classified in this group (Fig. 3B). No indicator species were found for this group, meaning that no species were exclusively present in this group nor present in all of the sites of the group. The dry *Sphagnum* group ($n = 12$) also exhibited a high *Sphagnum* species cover (approximately 60%) characterized this time by a higher presence of hummock species (*S. fuscum*) associated with a higher presence of ericaceous shrubs particularly dominated by *Kalmia angustifolia*, evidence of drier conditions (Table 2). Most of the reference sites were grouped in this group (Fig. 3B). *Kalmia angustifolia*, *Rhododendron groenlandicum*, and *Picea mariana* were indicator species of the dry *Sphagnum* group (Table 3) and their cover was at least two times higher than in any other group. Lichens, mainly *Cladonia* sp., had the highest cover in this group and was only present (<1%) in other groups (Table 2).

Influence of Harvesting and Environmental Conditions on Vegetation

The selected harvesting, hydrological, and meteorological variables (see Table S2 for all selected explanatory variables) explained 77% of the variability of the vegetation surveyed in the 25 donor sites (Fig. 4) and all groups of explanatory variables had a significant influence. The harvesting group explained most of the variability (21%), followed by the time since harvesting (17%), the meteorology of the growing season following harvesting (9%) and the hydrology (3%) (independent variance; Fig. 5). The first gradient of the RDA separated the plots along a gradient of *Sphagnum* moss cover (RDA1; Fig. 4).

Bare and disturbed donor sites were more rectangle shaped, harvested with a rototiller, during the summer, adjacent to peat fields currently extracted and/or natural areas, and experienced a warm and dry first summer (Tsum and Max noP; Fig. 4). Wet *Sphagnum* donor sites were more common when there were more precipitations during the first summer after the harvesting (Psum and Per rain; Fig. 4). Those sites were mostly harvested with a rotary harrow during spring and were adjacent to unrestored peat fields and anthropic land (mainly agriculture). They were generally far (>15 m; Fig. 4) from open active drainage ditches. In regeneration donor sites were distributed between the two previous groups. The dry *Sphagnum* group is not displayed in Figure 4, because this group includes only reference sites (Fig. 3B).

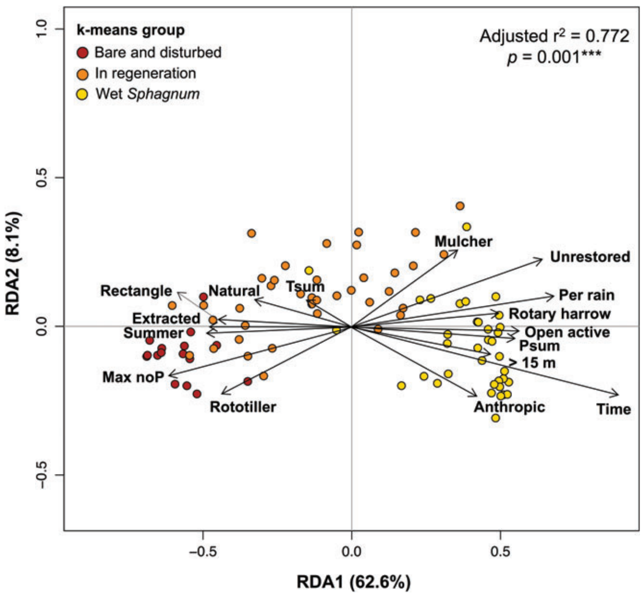


Figure 4. Redundancy analysis (RDA) triplot (scaling 1) with forward selected explanatory variables constraining the total cover of the dominant strata (peat and litter, disturbed substrate, graminoid plants, *Sphagnum* mosses, ericaceous species [not displayed]). To improve the visual clarity, only the most important explanatory variables were represented (spring, restored, Tspr, and A.I were selected but not represented on the RDA). The first axis of the RDA corresponds to a gradient of *Sphagnum* cover from low (left) to high (right). The site scores are represented as weighted averages of the explanatory data.

Discussion

Regeneration of the Vegetation: Species Abundance and Composition

This study demonstrated that harvesting *Sphagnum* diaspores for peatland restoration in North America was possible without impeding the natural ability of self-regeneration of peatland ecosystems. Even if the *Sphagnum* layer is expected to regenerate within a 10-year timespan, our results show that a change in plant community structure and composition should also be expected.

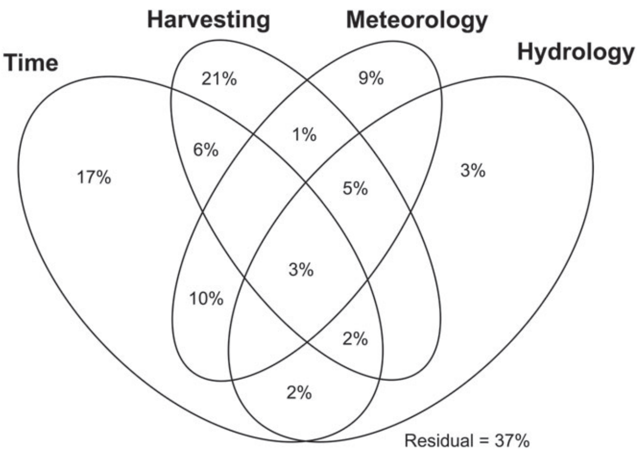


Figure 5. Venn diagram of the variation partitioning of the selected explanatory variables from each group (variable of time since harvesting, harvesting group [excluding time], meteorology group, hydrology group) carried out on the vegetation stratum matrix (see Table S2 for selected explanatory variables). Independent variances represent the variance explained by one group while the effects of other groups are controlled. Relative contributions of less of 1% were not represented.

For instance, trees and shrubs regenerated slowly in donor sites (Murray et al. 2017). Their regeneration, germination, survival, and growth are affected by the substrate water saturation (Chimner & Hart 1996; Hörnberg et al. 1997). Harvesting in donor sites lowers and flattens the surface, thus harvested donor sites become wetter (Murray et al. 2017) and do not exhibit the microtopography required for trees and shrubs germination. *Chamaedaphne calyculata* was the only ericaceous shrub regenerating more quickly; this species grew well in hollow habitats (Campbell & Rochefort 2001). On the contrary, *Kalmia angustifolia*, *Rhododendron groenlandicum,* and *Picea mariana* were associated with natural peatlands and preferred hummock habitats (Campbell & Rochefort 2001). These species might also have been over-represented in reference sites. As previously described, often all open parts of the peatland were harvested for donor material and the remaining adjacent nonharvested area left for natural reference inventory corresponded more to a treed *Sphagnum* peatland. This explained the higher cover of *K. angustifolium*, *R. groenlandicum,* and *P. mariana* in the dry *Sphagnum* group that are prevalent in drier forested *Sphagnum* peatland (Rydin & Jeglum 2013).

Eriophorum vaginatum has often been designated as a pioneer species (Buttler et al. 1996; Hughes & Dumayne-Peaty 2002) or invasive species (Tomassen et al. 2004; Lavoie et al. 2005) in disturbed peatlands. Because of its high wind-dispersal potential and relatively high fecundity, it is the most common herbaceous species in unrestored and restored extracted peatlands (Salonen 1987; Campbell et al. 2003) and in donor sites (Murray et al. 2017). In a previously extracted and unrestored peatland, Lavoie et al. (2005) found that *E. vaginatum* extent increases drastically with the rise of the water level and started to decrease 9 years later. In successfully restored peatlands, *E. vaginatum* cover reached almost 40%, 5–6 years after the restoration and decreased afterward (Rochefort et al. 2013).

As found by Poulin et al. (2013), González et al. (2013), and Karofeld et al. (2016), *S. rubellum* was more abundant in regenerating peatlands than in natural peatlands. This species has a wide ecological amplitude (Gignac 1992) and is competitive when environmental conditions are variable (Pouliot 2011). The presence or absence of the other cryptogams (bryophytes and lichens) in donor sites is reflected by their habitat preference. For example, *Cladonia* species preferred drier habitat (Jasieniuk & Johnson 1982) and donor sites are usually selected to contain few lichens, and therefore, *Cladonia* sp. could be over-represented in natural sites.

Indeed, species that prefer wetter habitats (e.g. *C. calyculata*), pioneer (e.g. *Eriophorum vaginatum*), and competitive species (e.g. *S. rubellum*) were more abundant in regenerating donor sites while species with a slow growth rate (trees, shrubs, and lichens) were more abundant in reference sites. This highlights that the regeneration of the plant species is related to their life-history strategy and habitat niche. Reference sites were probably older in terms of successional age which might explain differences in species composition and abundance between donor and reference sites.

Influence of Harvesting: Time Versus Harvesting Conditions

Sphagnum regeneration in donor sites was linearly related to the number of years postharvesting. Indeed, all donor sites aged of 10 years or more had a similar *Sphagnum* cover to the reference sites. Faster regeneration was observed (2–5 years) in the oldest restoration projects (Rochefort et al. 1997; Rochefort & Campeau 2002). The harvesting of the plant fragments and other diaspores in those studies was carried out during spring with a bulldozer scraping the surface of peat mosses just above the ice. This method created little disturbance and harvested only the top 10 cm of the *Sphagnum* layer but requires specific conditions similar to the operation of northern Canadian winter roads over peatlands (snow is compacted in early winter to allow better penetration of the frost into the soil). However, this harvesting method is not used anymore because it requires the preparation described previously, which is time-consuming. Contrasting these earlier studies with our results, it suggests that a winter harvesting above frozen peat inducing low levels of soil disturbance can speed up *Sphagnum* regeneration, showing that harvesting practices might be more important than time since harvesting.

Indeed, harvesting variables were the most influential factors on the recovery of the *Sphagnum* carpet in our study rather than time since harvesting. In the case of donor sites, this result means that choosing appropriate harvesting practices could accelerate *Sphagnum* regeneration. For example, the use of a rototiller and/or harvesting during the summer could create more ruts and reduce the overall wetness of the donor sites by preventing even water distribution (Price et al. 1998). The use of less disruptive methods such as a rotary harrow or a forest mulcher and/or harvesting when the soil deeper layers are frozen is thus recommended. However, it is important to note that the season of harvesting does not always ensure that the soil is frozen and the presence of a thick layer of frozen peat should be validated before harvesting. Contrary to what is recommended by Quinty and Rochefort (2003), harvesting in long and narrow strips seemed to have a negative influence on *Sphagnum* regeneration. Though, this factor might have been misevaluated, because of the difficulty to evaluate the shape of the donor area long after harvesting. Overall, disturbances of donor sites should be kept to a minimum by minimizing the exposure of bare peat, sinking the machinery, and creating ruts. To do so, the choice of adequate donor sites might be the solution, as the choice of other harvesting practices. Indeed, donor sites that are too wet should be avoided or only harvested when the soil is frozen. However, choosing drier donor sites to avoid machinery sinking might reduce *Sphagnum* regeneration after harvesting and the quality of the restoration material. This latter recommendation applies if these drier donor sites are dominated by trees, shrubs (e.g. *K. angustifolia*), forest mosses (e.g. *Pleurozium scheberi* and *Dicranum* spp.), lichens and present a low *Sphagnum* content.

In terms of *Sphagnum* regeneration, selecting a donor site close to currently extracted peat fields was not optimal because of surface drainage in extracted peat fields (Landry & Rochefort 2012) and probably also of aerial peat deposition by wind on regenerating *Sphagnum* mosses (Faubert & Rochefort 2002). However, in terms of logistical and practical constraints, these donor sites adjacent to extracted fields should not be overlooked because they represent cheap and easily accessible propagules for restoration. The proximity of natural areas surprisingly decreased *Sphagnum* regeneration. Even if it can provide a source of diaspores for recolonization, natural areas were mainly composed of forests, which lowered water table levels by evapotranspiration (Ahti & Hökkä 2006). Even if donor sites were surrounded by intact peatlands, *Sphagnum* mosses have a low potential of recolonization by spores (Campbell et al. 2003). The presence of unrestored peatland fields which often implies inactive ditches that filled up over time had a positive influence on *Sphagnum* regrowth. Also, wind erosion of peat is reduced in unrestored peatland fields with the formation of a surface crust (Campbell et al. 2002).

Influence of the Environmental Conditions

Meteorological factors also had a significant contribution in explaining *Sphagnum* moss recovery. As found by Chirino et al. (2006), warm temperatures and prolonged drought during the first summer negatively affected the recolonization by mosses. Reduced air humidity decreased the desiccation resilience potential of *Sphagnum* mosses (Sagot & Rochefort 1996). On the other hand, high precipitation well-distributed during the first summer positively influenced *Sphagnum* regeneration, agreeing with Chirino et al. (2006) and Backéus (1988).

Hydrology had a less important influence than expected, probably because the proxies chosen were not appropriate or misestimated, especially for the evaluation of the type of drainage ditch using satellite images. The use of the adjacent area might be a better proxy for the evaluation of hydrology. Also, unlike restored peatlands, donor sites still have an acrotelm that increases soil wetness and reduces water tension (Price et al. 2003). Because of the presence of an acrotelm and the peat properties (e.g. low hydraulic conductivity), the influence of drainage on donor sites might have been minor.

In conclusion, this study is the first to our knowledge to demonstrate that harvesting of *Sphagnum* propagules in a natural peatland for ecosystem restoration is not damaging to peatland ecosystems. (1) We showed that the *Sphagnum* layer should regenerate back to covers similar to natural peatlands within an approximate period of 10 years which is relatively rapid compared to Silvan et al. (2017) (30 years) and Elling and Knighton (1984) (20 years). Species abundance and composition varied between donor sites and natural peatlands. Pioneer, competitive, and preferential species of wet microhabitat were more abundant in donor sites. (2) Furthermore, in order to improve the regeneration of donor sites, the skills of the practitioner are an essential ingredient, because harvesting practices are the most influencing variables for vegetation regeneration. (3) Prolonged drought during the first summer following harvesting impeded donor site recovery and the importance of hydrologic conditions on vegetation regeneration was underestimated due to the wrong choice of proxies. Finally, donor sites could even be used as "nurseries"—that is, to use them multiple times for subsequent restoration projects—but more research will be needed on the impact of cyclical harvesting on *Sphagnum* regeneration potential.

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Supporting Information

The following information may be found in the online version of this article:

Figure S1. Disposition of natural reference site plot in the case of the presence and absence of an adjacent open *Sphagnum* peatland.

Figure S2. Pictures of the vegetation plot in the (a) bare and disturbed (b), in regeneration (c), wet *Sphagnum*, and (d) dry *Sphagnum* sites.

Table S1. Characteristics of the 25 studied donor sites.

Table S2. Explanatory variables selected during the forward selection procedure for the RDA done on the vegetation stratum matrix for all the donor sites.

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